

Liquid-Liquid Extraction

Final Lab Report
Unit Operations Lab 5
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1. Executive Summary

The objective of the experiment is to test the functionality of a three-stage liquid-liquid extraction system in the solvent removal of acetal acid in white mineral oil using water as the extracting solvent. The impact of the agitator speed on the extraction efficiency and the performance of the stage was also examined to identify the best operating conditions.

The three-stage mixer-settler system was used to conduct the experiment under the conditions of countercurrent flow. A white oil feed mixture with a concentration of approx. 1 wt% acetic acid was in contact with water, and the system operated at three agitator speeds (390, 600, and 775 RPM). At a steady state, samples at each stage were collected, and titration was used to determine the concentration of acetic acid in extract (water) and raffinate (oil) phases.

The results suggest that the speed of agitators influences the performance of extraction significantly. As the results of plots indicate, agitation of the mixture (600 RPM) had a positive effect on the transfer of acetic acid to the water phase, especially in the initial stage, where the largest concentrations were noted to have taken place. Reduced agitation rates showed decreased mass transfer and inefficient extraction. Across all the cases, the concentration of acetic acid declined with the rise in the stage number, which supports the fact that staged extraction is more effective in enhancing separation.

It was concluded that agitation increases mass transfer by increasing phase contact, but deviation of ideal behavior of equilibrium indicated that incomplete phase separation, and limitation of mixing influences system performance. Overall, the experiment proves that a multi-stage extraction system is suitable to achieving the reduction of solutes concentration, and that the rate of agitators is one of the most important variables to take into consideration when optimizing the extraction rate.

2. Introduction

This experiment aimed at studying the extraction of acetic acid in white mineral oil with water in a three-stage liquid-liquid extraction system and determining the influence of the agitator speed on the separation efficiency. Liquid-liquid extraction is a significant separation method applied in chemical processing industries, pharmaceutical industries, and environmental engineering, especially in cases where distillation is not possible and effective.

A three-stage mixer-settler system was applied in this experiment, and each stage was made up of a mixing zone and a settling zone. In mixing, the two immiscible phases of liquids are subjected to close contact, which facilitates the transfer of acetic acid in the oil stage to the water phase. The phases are then separated by density differences because of the mixing, and extract and raffinate streams can be collected. The degree of separation is determined by the relationships of equilibrium between the two phases and the efficiency of mass transfer.

The system was run at various agitator speeds (390, 600 and 775 RPM) with a constant flow rate to control the effect of mixing intensity. Titration provided a result in which the

acetic acid concentrations in the two phases at each stage were measured and used to assess system performance. Such results were compared to the expected equilibrium behavior and examined through the concepts of stage efficiency and multistage extraction.

3. Theory (10 pts)

Liquid-liquid extraction is a mass transfer operation used to separate a solute from one liquid phase into another immiscible liquid phase based on differences in solubility. In this experiment, acetic acid was transferred from a nonpolar carrier phase (white oil) into a polar solvent phase (water). The driving force for this separation was the difference in chemical potential of the solute between the two liquid phases, which resulted in the redistribution of acetic acid until equilibrium was approached (Bjørnstad, 2018).

At equilibrium, the concentration of solute in the extract phase () and raffinate phase () are related through an equilibrium distribution curve. This relationship is often expressed using a distribution coefficient or partition coefficient, which is defined as the ratio of solute concentration in the solvent phase to that in the feed phase at equilibrium:

For dilute systems such as acetic acid in water and oil, this relationship can often be approximated as linear over a limited range:

Where m represents the slope of the equilibrium line (Subramanian, 2021). The equilibrium data provided in the lab manual (Appendix I) enabled comparison between experimental results and ideal equilibrium behavior (UIC, 2025).

The extraction process in this experiment was carried out in a three-stage mixer-settler system, where each stage consist of a mechanically agitated mixed zone followed by a gravity-based settling zone. In the mixer, intense agitation disperses one liquid phase into the other, increasing interfacial area and enhancing mass transfer. The rate of mass transfer depends on interfacial area, diffusion coefficients, and concentration gradients, and is often described using film theory or two-film theory. According to this model, resistance to mass transfer exists in thin boundary layers on either side of the interface, and the overall transfer rate is governed by the slower of the two phases (Green & Southard, n.d.).

After mixing, the dispersion enters the settler where the two phases disengage due to density differences. Efficient phase separation is critical, as incomplete settling can lead to entrainment and inaccurate composition measurements. For multistage countercurrent extraction, mass balances are used to describe solute transfer between stages. The operating line for the system can be expressed as:

Where F and S are the feed and solvent flow rates, respectively. This relationship is used in conjunction with equilibrium data in the McCabe-Thiele method to determine the number of theoretical stages required for separation (McMaster University, n.d.). The performance of each stage depends on how closely the exiting streams approach equilibrium. In practice, real stages do not achieve complete equilibrium, so stage efficiency is used to quantify performance. The Murphree stage efficiency is defined as the ratio of the actual change in solute concentration to the change that would occur if equilibrium were reached (McMaster University, n.d.):

Where y^* represents the equilibrium concentration corresponding to x . The efficiency provides a measure of how closely each stage approaches ideal equilibrium conditions.

To analyze multistage extraction systems, the McCabe-Thiele graphical method is commonly used. This method uses equilibrium data along with an operating line, which is derived from overall mass balances, to determine the number of theoretical stages required for a given separation (McMaster University, n.d.). In this experiment, the measured concentrations of acetic acid in each stage were plotted and compared with equilibrium data to estimate the number of ideal stages and evaluate system performance.

Agitator speed plays a key role in the performance of the extraction process. Increasing mixing intensity enhances mass transfer by increasing interfacial area and reducing boundary layer thickness, but excessive agitation can lead to stable emulsions that hinder phase separation. Therefore, an optimal range of agitation must be maintained to balance mass transfer efficiency and phase disengagement (Bjørnstad, 2018). In this experiment, agitator speed was varied while keeping flow rates constant to isolate its effect on extraction efficiency and stage performance.

Experimentally, the concentration of acetic acid in both phases is determined using acid-base titration with sodium hydroxide and phenolphthalein indicator. The concentration was calculated using:

This method relies on stoichiometric neutralization of acetic acid and provides a quantitative measure of solute concentration. These measured values are then used to construct equilibrium plots, determine stage efficiencies, and compare experimental performance with theoretical predictions (UIC, 2025).

4. Results (15 pts)

Figure one is in regard to the acetic acid weight percent within the respective water and oil samples (extract and raffinate samples).

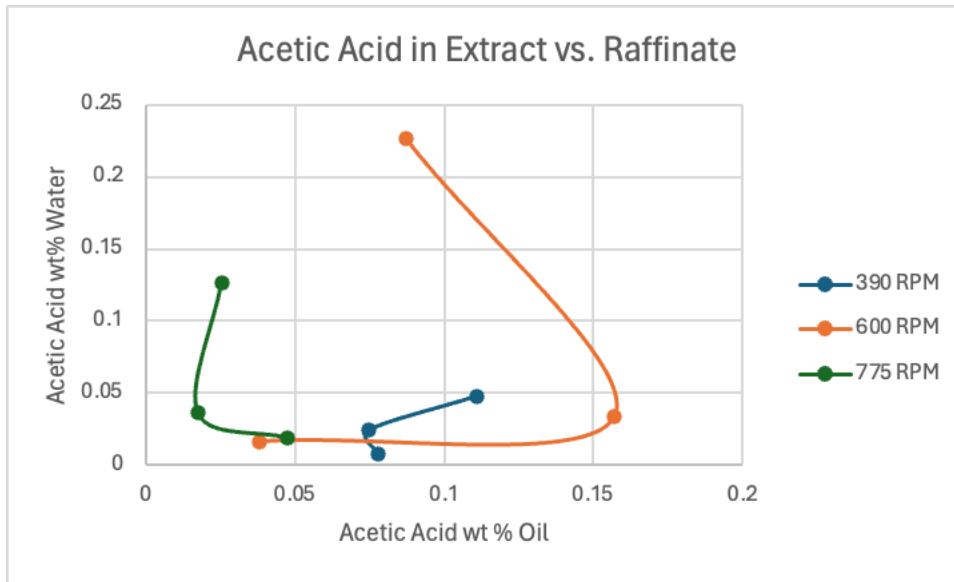


Figure 1: A plot of the acetic acid in weight percent within the mineral oil and the water at the same stage, for varying agitator speeds.

Figures 2 and 3 are made with regard to the acetic acid concentration in mol/L at each of the three stages at the varying agitator speeds. Note water samples and oil samples taken at the same stage for one speed were taken at approximately the same speed.

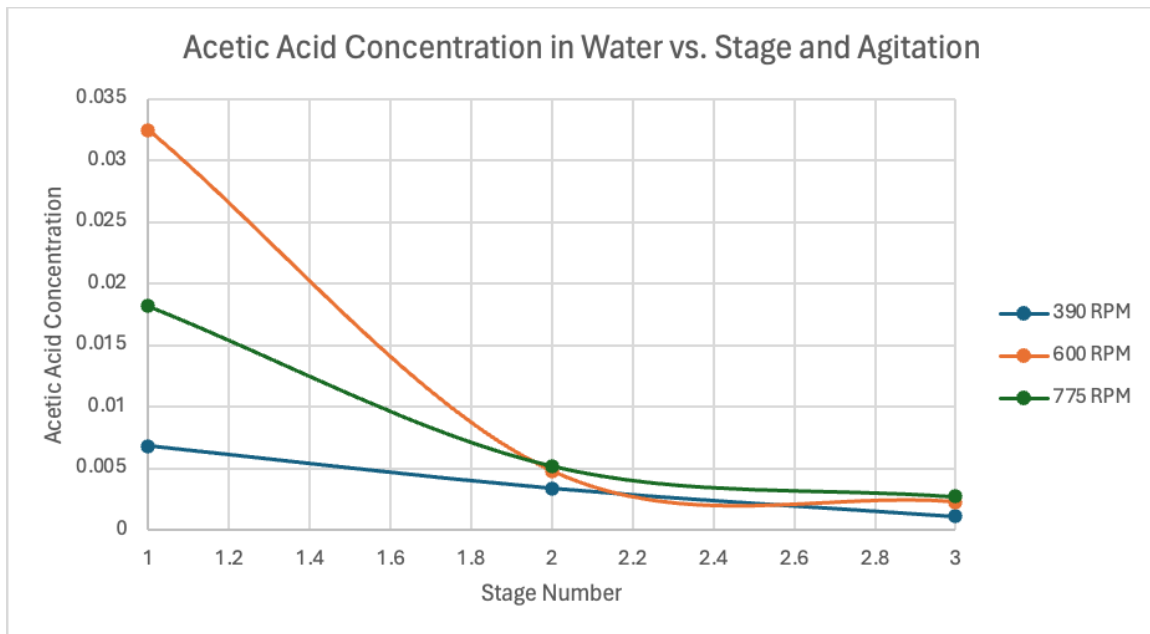


Figure 2: The concentration in mol/L of acetic acid within the water at each of the three stages at different agitator speeds.

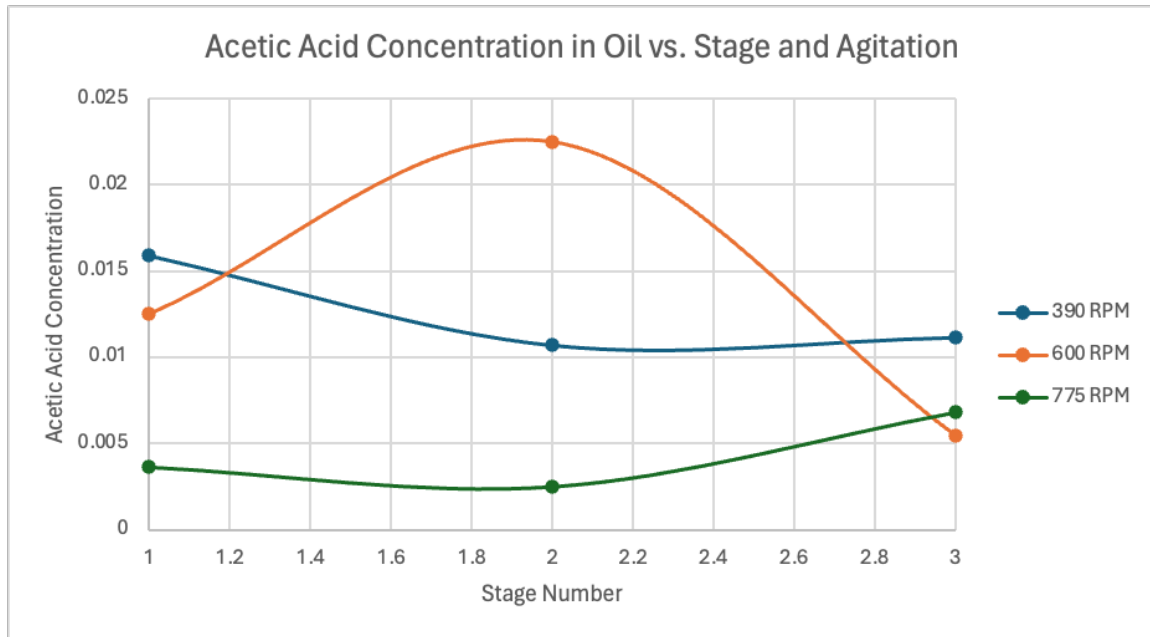


Figure 3: The concentration in mol/L of acetic acid within the white mineral oil at each of the three stages at different agitator speeds.

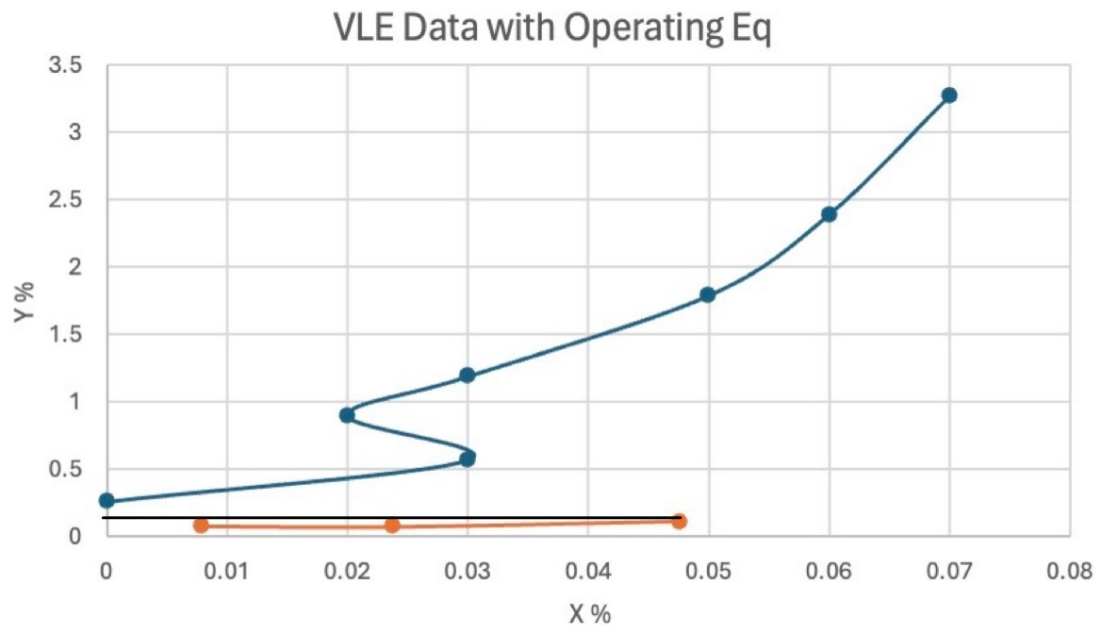


Figure 4: Values from agitator speed 390 RPM X and Y plotted against VLE data provided within the lab manual.

Figure 4 was meant to complete the McCabe-Thiele stages, however, note no true stages could be found from this plot, and therefore no realistic efficiency could be solved for.

5. Discussion (15 pts)

The experimental results generally agree with the theoretical principles governing liquid-liquid extraction. In a multistage countercurrent extraction process, the concentration of

solute in the raffinate phase is expected to decrease as the mixture progresses through each stage, while the extract phase becomes enriched with solute (McMaster University, n.d.). The data collected in this experiment follow this expected trend. Across all agitator speeds, the concentration of acetic acid in the oil phase decreased between stages, indicating that mass transfer from the oil phase into the aqueous solvent occurred as intended. This behavior is consistent with mass transfer theory, which states that solute redistribution between immiscible phases occurs until the system approaches equilibrium governed by the distribution coefficient (Bjørnstad, 2018). The stagewise decrease in raffinate concentration also reflects the benefit of staged extraction systems, where each stage provides an additional opportunity for solute transfer and moves the system closer to equilibrium conditions (Chemical Engineering Handbook, n.d.).

The influence of agitator speed on extraction performance was clearly observed in the experimental results. At low agitation (390 RPM), the concentration of acetic acid transferred to the water phase remained relatively small, suggesting that insufficient mixing limited the available interfacial area between the two phases. According to mass transfer theory, the rate of solute transfer depends strongly on the interfacial area and the thickness of the diffusion boundary layers surrounding dispersed droplets (Subramanian, 2021). Lower agitation intensity results in larger droplets and reduced phase contact, which decreases the rate of diffusion across the liquid–liquid interface. In contrast, increasing the agitation speed to 600 RPM significantly improved extraction performance, particularly in the first stage where the concentration of acetic acid in the water phase was highest. This result is consistent with literature findings indicating that moderate agitation enhances droplet dispersion and interfacial surface area, thereby increasing mass transfer rates between phases (Bjørnstad, 2018).

Interestingly, the highest agitator speed tested (775 RPM) did not produce the greatest extraction efficiency. While increased agitation typically improves mass transfer, excessively high mixing speeds can create operational issues such as stable emulsions or incomplete phase disengagement in the settler (Chemical Engineering Handbook, n.d.). These conditions reduce the effective separation between the extract and raffinate streams and can lead to entrainment of one phase within the other. The experimental data suggest that this may have occurred at the highest mixing speed, where the concentration of acetic acid in the water phase was lower than that observed at 600 RPM. Similar behavior has been reported in liquid–liquid extraction studies, where an optimal mixing intensity exists that balances droplet dispersion and efficient phase separation (Bjørnstad, 2018). Therefore, the results support the conclusion that an intermediate agitation rate provides the most effective extraction performance for this system.

Comparison of the experimental results with theoretical equilibrium behavior also provides insight into stage performance. Ideally, the compositions of the extract and raffinate streams leaving each stage would lie directly on the equilibrium distribution curve defined by the partition coefficient relationship between phases (McMaster University, n.d.). However, the experimental values did not perfectly match the equilibrium data, indicating that the real stages did not reach complete equilibrium. This deviation is expected in practical extraction systems because mass transfer resistance,

finite residence time, and imperfect mixing limit the extent to which equilibrium can be achieved within each stage. As described in mass transfer theory, the difference between the actual and equilibrium concentrations can be characterized using stage efficiency concepts such as the Murphree efficiency (McMaster University, n.d.). The observed deviations from equilibrium therefore suggest that the stages operated at efficiencies lower than 100%, which is typical for laboratory scale mixer-settler equipment.

Several experimental limitations may also explain discrepancies between the measured values and theoretical expectations. Incomplete phase separation in the settler sections could have resulted in entrainment of small droplets, which would affect the measured concentrations in both phases. Sampling errors may also have occurred during withdrawal of the oil and water samples, particularly if residual dispersion remained in the collection region. In addition, the determination of acetic acid concentration relied on acid-base titration using sodium hydroxide and phenolphthalein indicator, which introduces potential uncertainty related to endpoint detection and small measurement errors in titrant volume. These sources of experimental uncertainty are common in liquid-liquid extraction experiments and can lead to deviations from ideal equilibrium behavior (UIC Liquid Extraction Lab Manual, 2025).

Despite these limitations, several reliable conclusions can be drawn from the experiment. The data demonstrate that staged liquid-liquid extraction is an effective method for removing acetic acid from a nonpolar oil phase using water as the extracting solvent. The results also confirm that agitator speed plays a critical role in determining extraction efficiency by influencing droplet formation, interfacial area, and phase contact time. Among the conditions tested, the intermediate agitation rate of 600 RPM provided the most favorable balance between mass transfer enhancement and effective phase separation. More speculative conclusions involve the exact stage efficiencies and equilibrium deviations, which would require additional measurements and more precise equilibrium data to quantify accurately.

The findings of this experiment have broader implications for the design and operation of industrial extraction processes. Mixer-settler systems are widely used in chemical processing industries, and understanding the relationship between mixing intensity and mass transfer performance is essential for optimizing separation efficiency while minimizing energy consumption and operational problems such as emulsification (Chemical Engineering Handbook, n.d.).

Future investigations could further explore these effects by measuring droplet size distributions, determining stage efficiencies quantitatively, or evaluating the effect of solvent to feed ratio on extraction performance. Additional experiments could also compare the experimental results with predictions obtained using McCabe-Thiele analysis to better estimate the number of theoretical stages required for the desired separation.

Overall, the experimental results support the fundamental principles of liquid-liquid extraction and demonstrate how operating conditions such as agitator speed influence mass transfer and stage performance in a multistage extraction system.

6. References (10 pts)

Liquid-Liquid Extraction Lecture Notes. (n.d.). McMaster University.
https://learnche.org/wiki_4M3/images/b/b7/2013-4M3-Liquid-Liquid-Extraction.pdf

Mass Transfer Operations: Extraction Notes. (2021).
<https://msubbu.in/ln/pce/PCE-Lecture-4-5-MassTransfer-Extraction.pdf>

Bjørnstad, T. (2018). *Introduction to liquid-liquid extraction*. University of Oslo.
<https://www.researchgate.net/publication/354814096>

Chemical Engineering Handbook: Liquid-Liquid Extraction Chapter. (n.d.).
<https://thunderbooks.wordpress.com/wp-content/uploads/2009/05/chemenghbk-ch15-liquid-liquid-extraction.pdf>

UIC Liquid Extraction Lab Manual. (2025). *Liquid-liquid extraction using a three-stage mixer-settler*. University of Illinois Chicago.

7. Appendix I: Data Tabulation/Graphs (10 pts)

Table 1- Initial data collected to find the starting concentration of acetic acid in mineral oil.

Sample Volume (mL)	NaOH Conc (N)	NaOH Used (mL)	Acetic Acid (mol/L)	Acetic Acid (g/L)
22	0.05	2.3	0.00523	0.314

Table 2- Raw data collected from experiment. It includes all stages and all RPMs. For example; the oil sample is then linked to the "NaOH used-Oil".

Agitator Speed (RPM)	Stage	Oil Sample (mL)	Water Sample (mL)	NaOH Used – Oil (mL)	NaOH Used – Water (mL)
390	1	22	22	7.0	3.0
390	2	22	22	4.7	1.5
390	3	22	22	4.9	0.5
600	1	22	22	5.5	14.3
600	2	22	22	9.9	2.1
600	3	22	22	2.4	1.0
775	1	22	22	1.6	8.0
775	2	22	22	1.1	2.3
775	3	22	22	3.0	1.2

8. Appendix II: Error Analysis (10 pts)

The accuracy and reliability of the experimental results were evaluated through both quantitative uncertainty analysis and qualitative assessment of potential error sources. While the overall trends observed in the extraction system were consistent with theoretical expectations, several sources of uncertainty likely influenced the precision of the measured concentrations and calculated performance metrics.

The primary source of quantitative uncertainty in this experiment was from the titration procedure used to determine acetic acid concentration. Uncertainty in this calculation was primarily influenced by measurement error in NaOH volume and sample volume. Assuming a burette uncertainty of ± 0.05 mL and a sample volume uncertainty of ± 0.05 mL, the propagated relative uncertainty in concentration can be estimated. For a typical titration, this corresponds to an approximate uncertainty of ± 1 -3% in calculated concentration.

To further quantify variability, replicate data across the three stages at each agitation speed were analyzed. For example, at 390 RPM and the oil phase, the measured concentrations were 0.0159 mol/L, 0.0107 mol/L, and 0.0111 mol/L, this yields a mean of ~ 0.0126 mol/L:

With an estimated standard deviation of ~ 0.0027 mol/L:

This corresponds to a relative standard deviation (RSD) of approximately 21%:

This relatively high variability suggests that uncertainty is dominated not only by titration error but also by process-related factors such as mixing and sampling.

Several experimental factors likely contributed to deviations between experimental results and ideal behavior. One of the most significant sources of error was incomplete separation of the oil and water phases in the settler. At higher agitation speeds, increased dispersion likely led to entrainment of one phase into the other, artificially altering measured concentrations. This is supported by irregular trends observed in the extract vs. raffinate plot, where data points do not perfectly follow a smooth equilibrium curve.

Samples were taken from each stage while the system was operating. If sampling occurred before full disengagement, or if the sampling point captured a mixed region, the measured composition would not represent the true bulk phase. Even small amounts of

cross-contamination can significantly impact titration results due to the relatively low concentrations involved.

The use of phenolphthalein introduces subjectivity in identifying the endpoint, especially for dilute samples. Additionally, titration of the oil phase required adequate mixing with the aqueous NaOH, and insufficient mixing could lead to incomplete neutralization and underestimation of acid concentration.

The experiment relied on maintaining a constant solvent-to-feed ratio. Small fluctuations in rotameter readings or pump performance would affect the operating line and shift the system away from ideal conditions.

Despite these sources of error, the overall trends in the data remained physically reasonable. However, the scatter in the equilibrium plot and variability between stages indicate that quantitative agreement with theoretical equilibrium and ideal state performance was limited. As a result, calculated values should be interpreted as approximate rather than exact.

Several improvements could enhance the accuracy and reliability of the experiment. Increase settling time to ensure complete phase disengagement before sampling. Use replicate titrations for each sample to reduce analytical uncertainty. Implement more precise volumetric equipment such as digital burettes. Improve mixing control, particularly at high RPM, to avoid emulsification. Verify steady-state conditions by monitoring outlet concentrations over time before sampling. Finally, use phase separators and coalescers to reduce entrainment between phases.

9. Appendix III: Sample Calculations (10 pts)

For calculating the initial amount of acetic acid needed to add, the following math was done based on the need of 2.3mL of 0.05 M NaOH being needed to neutralize 22mL of white mineral oil with acetic acid sample:

For calculating the weight percent of acetic acid, the follow calculations were done for stage one, 390 RPM, oil sample; given that the amount of oil sampled is 22mL, and the concentration of the NaOH is 0.05M:

The weight percent was then approximated using the assumption that the majority of the mass came from the white mineral oil,

10. Appendix IV: Individual Team Contributions

NAME: Kendall

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	3	Collected samples from the different stages, set up flow rates, and set agitator speeds.
Pre-Lab Editing	0	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	1	Color coded and labeled the apparatus system
Materials & Supplies (prelab)	0.5	Listed out all of the materials required for the lab, many involved the sample collection and testing
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0.5	Filled out table regarding safety and hazards
Final Lab Editing	0	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	4	Calculated values from raw data and made the corresponding graphs.
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0	

Data Tabulation/Graphs (final)	1	Setting up the initial spread sheet and performing calculations.
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	2	Explaining how calculations were performed, reflected in results and data tabulated in appendix one.
Power Point Presentation	0	
Total	12	

NAME: Lindsey

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	3	Gathered information on prelab day and then helped collect data, record numbers, and prepare flasks for sampling.
Pre-Lab Editing	0.5	Spell check and formatting
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	2	Wrote out a step by step procedure to follow during the lab period.
Project Safety Evaluation (prelab)	0	
Final Lab Editing	0.5	Spell check and formatting
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	3	Used multiple sources to outline the theory behind the experiment and important equations that were used.
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	0.5	Formatted all sources.
Data Tabulation/Graphs	0	

(final)		
Error Analysis (final lab)	2	
Sample Calculations (final Lab)	0	
Power Point Presentation	1	Transferred information from the report to the presentation.
Total	12.5	

NAME: Christian

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	3	Very involved in every part of the laboratory
Pre-Lab Editing	.5	Ensured all necessary topics were included
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)	.5	Considered all the possible safety hazards
Final Lab Editing	.5	Made sure all our sources included and that they were in the correct style
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)	2	Compared our findings to literature. Pensive about the future work that could be done
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	.5	Organized our raw data into easily digestible tables
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	Organized the PowerPoint

		and focused on safety and procedure
Total	8	

NAME: Srija

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	2	Helped with the experiment and collected the data for the results
Pre-Lab Editing	1	Did the prelab
Experimental Objective (Prelab)	1	Did the objective
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	3	Did the introduction and the executive summary
Executive Summary (final lab)	1	
Introduction (Final lab)	1	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	0	
Conclusion (final lab)	0	
References (final and/or prelab)	1	Added one of the references
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	2	Transferred the information to the powerpoint
Total	12	